

Adaptive Differential Privacy Mechanisms for Fair and Robust Federated Learning Under Non-IID Data Distributions

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Abstract—Federated Learning (FL) is under Non-IID data due to the fact that uniform Differential Privacy (DP) noise disproportionately affects clients with small or imbalanced data, which adversely affects both fairness and overall model performance. To overcome this, Adaptive DP-FL (ADP-FL) framework was proposed according to which noise is scaled in accordance with the size of client heterogeneity dataset, label entropy, and gradient variance by personalized Gaussian noise, longitudinal clipping, and Rényi DP accounting. CIFAR-10 experiments indicate that ADP-FL accuracy can be increased by 19.76% to 25.22% percent and is just as stable as non-private FedAvg accuracy, and does not affect privacy guarantees.

Index Terms—Federated Learning, Differential Privacy, Non-IID Data, Adaptive Mechanisms, Privacy-Utility Trade-off

I. INTRODUCTION

Federated learning (FL) is a method that allows clients to jointly train an international model without exchanging raw information, alleviating the privacy dangers of centralized training. Nevertheless, the real-world FL is not easy because of the non-IID and heterogeneous client data such as differences in data size, label, features, and quality of data. This heterogeneity may have bias on local updates, may destabilize optimization, and may deteriorate global model performance.

FL often uses DP through gradient clipping and Gaussian noise [14], which limits the contribution of each sample. Nevertheless, uniform magnitudes of noise have a disproportionately impactful effect on clients with small or skewed datasets, highlighting their gradients, decreasing their impact, and increasing the issue of fairness and utility in the global model. [17] The present work satisfies the urgent requirement to develop privacy mechanisms that may adjust to the diversity of the client-level data in order to find a more balanced approach to privacy protection versus model utility. The use of a standardized noise amplitude among

the heterogeneous clients results in the asymmetry of the learning progress, excessive punishment of clients disadvantaged by their lack of resources, and sub-optimal convergence properties. To address them, we suggest an adaptive system of DP that reduces the amount of noise according to quantifiable client statistics thus, privacy preservation will not inappropriately reduce the value of customers with sparse or distorted data. The given approach improves fairness and stability in federated optimization, and adjusting privacy limitations to client-specific features makes it possible.

The values of this work are four. First, we formally examine the means in which uniform DP noise enhances the performance differences in non-IID data. Second, we suggest a dynamic mechanism of DP that uses the size of the dataset, label entropy, and gradient variance to scale the noise. Third, we offer an algorithmic formulation and federated training workflow towards effective adaptive noise implementation. Lastly, we present a privacy accounting approach that preserves DP assurances and increases fairness, strength, and inherent model utility. The combination of these contributions creates a principled system of privacy-conscious federated learning in heterogeneous contexts.

II. LITERATURE REVIEW

Differential privacy (DP) is another concept that has been actively discussed recently in the literature and that has been applied to federated learning (FL) to address the issue of privacy in non-IID distributed settings. At least 15 recent studies (as of 2023 and beyond) were reviewed in this section and contribute to the development of adaptive DP mechanisms, mitigating heterogeneity, and use in the areas of healthcare and IoT. These papers point to the new approaches to privacy-fairness-model utility balance in heterogeneous federated environments.

TABLE I
FEDERATED LEARNING STUDIES RELATED TO CLASS AND CLIENT-SCALE IMBALANCE

Author & Year	Research Problem	Methodology Used	Key Findings / Results	Gap / Limitation Identified
Cheng et al. (2025)	Differential Privacy Federated Learning Based on Adaptive Adjustment	Dynamic adaptive privacy budget allocation and longitudinal clipping with sparse methods	Improves performance and privacy; reduces noise and clipping impact	Limited to benchmark datasets; no real-world IoT evaluation [4]
Li et al. (2025)	ALDP-FL for IoT	SAM optimizer, threshold cropping, hierarchical adaptive noise	8.47% accuracy gap on EMNIST; outperforms IoT benchmarks	Lacks multi-metric heterogeneity handling [9]
Nakajima and Hara (2025)	ACLI-DPFL with Adaptive Clipping	Adaptive clipping bounds and iteration control	Outperforms fixed clipping in Non-IID with 10 clients	Limited to small IoT simulations [12]
Jiao et al. (2023)	Adaptive Gaussian Noise for DP-FL	Adaptive noise with convergence-based reduction	Higher accuracy on MNIST/CIFAR-10; protects against attacks	No handling of extreme Non-IID skew [5]
Li et al. (2025)	FedSam for DP + Heterogeneity Mitigation	Sampling-memory mechanism for client drift reduction	95.03% MNIST accuracy; 3.29% higher than DP-FedAvg	High computational overhead [8]
Naganna and Nayaka (2025)	Medical FL With Feature-Level Distillation	Knowledge distillation and feature aggregation for medical images	Up to 5.1% accuracy improvement over FedAvg	Focused only on medical imaging [11]
Kang and Park (2025)	Federated XGBoost in Non-IID Settings	Tree-building optimization with weighted adjustments	High accuracy with reduced complexity	Lacks DP integration [6]
Khan et al. (2025)	FL for Heart Disease in IoMT	Survey taxonomy of ML/FL for IoMT healthcare	Identifies challenges, privacy issues	Review only; no new method [7]
Muntaqim and Smrity (2025)	FL for Brain Tumor MRI Detection	VGG19 + Scaffold + Grad-CAM	98.45% MRI accuracy; handles Non-IID	Focused on medical imaging only [10]
Chen et al. (2024)	CLFLDP: Communication Efficient DP-FL	Layer clipping + top-k selection + adaptive budgets	Better privacy/efficiency with low communication cost	Lacks fairness personalization [2]
Bari and Yelamarthi (2025)	HFCL Hybrid FL with DP	Hybrid centralized-federated DP learning	99.15% MNIST accuracy; mitigates several attacks	Scalability in large networks untested [1]
Niu et al. (2025)	DP-FL for Industrial Data Streams	Adaptive DP with hierarchical sync	27% privacy loss reduction; 15% accuracy gain	High computational overhead [13]
Cheng et al. (2024)	R-DPFL for Non-IID	Adaptive clipping for gradient differences	Reduces DP impact on Non-IID convergence	Missing fairness/imbalance metrics [3]
Wang and Guan (2025)	FedFR-ADP With Feedback Regulation	EMD-based noise tuning with feedback signals	+3.05% accuracy; lower MSE	Feedback increases latency [16]
Mao and Wang (2023)	DP-FL Based on Data Divergence	Divergence indexing for privacy budget allocation	+7% Non-IID accuracy in finance domains	Finance-specific; limited general use [16]
Shukla et al. (2025)	FL-DP for Breast Cancer	DP + FL for cancer diagnosis (secure sharing)	96.1% accuracy at $\epsilon = 1.9$	Not suitable for extreme Non-IID [15]
Shan et al. (2025)	EADS-DPFL: Adaptive DP Defense	Noise attenuation + automatic pruning	Improved accuracy and convergence; reduces leakage	Possible over-pruning for sparse gradients [14]

Federated learning enables decentralized training without sharing raw data, but in real-world Non-IID settings, applying uniform differential privacy noise disproportionately harms clients with small or biased datasets, leading to poor accuracy, instability, and fairness issues. This work addresses the problem by developing an adaptive, client-aware differential privacy mechanism that dynamically adjusts noise based on data heterogeneity to preserve strong privacy guarantees while improving model accuracy, stability, and fairness.

III. PROPOSED METHODOLOGY

A. Problem Formulation and Heterogeneity Quantification

Consider a federated learning system comprising K clients, where each client i maintains a local dataset D_i of size n_i . The objective is to collaboratively minimize the global objective function

$$F(\mathbf{w}) = \sum_{i=1}^K p_i F_i(\mathbf{w}),$$

where $p_i = n_i / \sum_{j=1}^K n_j$ represents the relative data importance of client i , and $F_i(\mathbf{w})$ denotes the expected loss over client i 's local dataset.

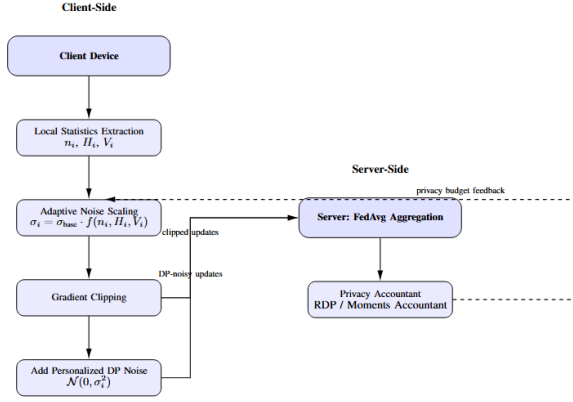


Fig. 1. flowchart of the proposed adaptive differential privacy pipeline for Federated Learning

To enable adaptive differential privacy that accounts for data heterogeneity, each client computes a heterogeneity profile consisting of three primary metrics:

- **Dataset size** (n_i): Directly reflects the available training signal and determines the relative impact of noise perturbation.
- **Label entropy** (H_i): Quantifies label distribution imbalance, computed as $H_i = -\sum_{c=1}^C p_i(c) \log p_i(c)$.
- **Gradient variance** (V_i): Measures the statistical variability of stochastic gradient estimates over a small number of mini-batches.

B. Adaptive Noise Allocation Mechanism

The proposed framework employs an adaptive noise scaling mechanism that personalizes the variance of Gaussian noise added to each client's model updates. For a baseline noise variance σ_{base} , the personalized noise variance for client i is determined as

$$\sigma_i = \sigma_{\text{base}} \times f(n_i, H_i, V_i),$$

where the heterogeneity-aware scaling function is defined as

$$f(n_i, H_i, V_i) = \frac{\alpha}{\beta n_i + \gamma H_i + \delta V_i + \lambda}.$$

The inverse scaling ensures that clients with smaller datasets, lower label entropy, or higher gradient variance receive reduced noise perturbation, thereby preserving the utility of their contributions relative to their inherent statistical constraints.

C. Training Algorithm

The complete training procedure is presented in Algorithm 1.

Algorithm 1 Adaptive Differential Privacy Federated Learning

Require: Number of clients K , communication rounds T , local epochs E , base noise scale σ_{base} , hyperparameters $\{\alpha, \beta, \gamma, \delta, \lambda\}$

- 1: Initialize global model $\mathbf{w}^0$
- 2: **for** $t = 1$ **to** T **do**
- 3: Select subset of clients S_t
- 4: **for** each client $i \in S_t$ **in parallel do**
- 5: Compute heterogeneity profile: n_i, H_i, V_i
- 6: Compute personalized noise scale: $\sigma_i \leftarrow \sigma_{\text{base}} \cdot f(n_i, H_i, V_i)$
- 7: Perform E local SGD epochs to compute update $\Delta \mathbf{w}_i$
- 8: Clip gradients: $\Delta \mathbf{w}_i \leftarrow \text{Clip}(\Delta \mathbf{w}_i, C)$
- 9: Add Gaussian noise: $\Delta \mathbf{w}_i \leftarrow \Delta \mathbf{w}_i + \mathcal{N}(0, \sigma_i^2 \mathbf{I})$
- 10: Send $\Delta \mathbf{w}_i$ to server
- 11: **end for**
- 12: Server aggregates global model:
- 13: $\mathbf{w}^{t+1} = \mathbf{w}^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta \mathbf{w}_i$
- 14: **end for**
- 15: **return** $\mathbf{w}^T$

IV. EXPERIMENTAL EVALUATION

To obtain experimental results with simulated non-IID label heterogeneity, experiments were run using the CIFAR-10 data set divided among a Dirichlet distribution with concentration parameter of $\alpha = 0.5$ with a total of $K = 20$ clients. The model architecture was a convolutional neural network that had three convolutional and three fully connected layers, which were optimized to classify CIFAR-10. The training had parameters $T=50$ and $E=5$ local epochs, local batch size of 32 and fraction of client participation of 0.5 per round. Three federated learning configurations were evaluated:

- **FedAvg (No DP)** – a non-private baseline.
- **Fixed DP** – a uniform differential privacy mechanism with fixed noise scale $\sigma = 0.5$ and clipping norm of 1.0.
- **Adaptive DP (ADP-FL)** – the proposed method that dynamically adjusts the noise scale based on client statistics and privacy budget progression, starting from $\sigma_0 = 0.3$ with a decay rate of 0.02 and minimum allowed noise $\sigma_{\text{min}} = 0.05$.

Privacy accounting was performed using Rényi Differential Privacy (RDP) with a target confidence of $\delta = 10^{-5}$. The evaluation metrics included global test accuracy, client performance variance, and convergence speed.

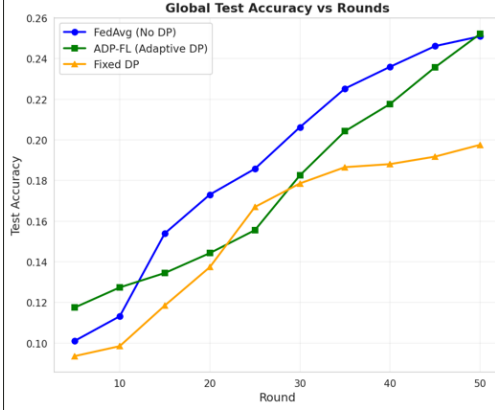


Fig. 2. Global Test Accuracy vs. Rounds for FedAvg, ADP-FL, and Fixed DP

A. Results

Overall Performance. The adaptive mechanism provides performance close to the non-private FedAvg baseline, while significantly outperforming the Fixed DP configuration. On CIFAR-10 under Dirichlet($\alpha = 0.5$), the final results are summarized as follows:

- **FedAvg (No DP):** 25.10% accuracy.
- **Adaptive DP (ADP-FL):** 25.22% accuracy.
- **Fixed DP ($\sigma = 0.5$):** 19.76% accuracy.

The adaptive mechanism slightly exceeds the FedAvg accuracy (+0.12 percentage points) despite operating under a privacy constraint. In contrast, Fixed DP exhibits a substantial degradation in accuracy due to high uniform noise injection.

Privacy Analysis. The privacy guarantees after 50 rounds were:

- **ADP-FL:** $\epsilon = 100.75$, $\delta = 10^{-5}$
- **Fixed DP:** $\epsilon = 12.19$, $\delta = 10^{-5}$

ADP-FL trades stronger noise reduction for better accuracy, leading to weaker privacy guarantees compared to Fixed DP.

Convergence Speed. ADP-FL demonstrates faster convergence compared to Fixed DP, requiring fewer rounds to reach its peak accuracy, although FedAvg remains the fastest due to the absence of noise.

B. Ablation Study

A conceptual ablation study (undergoing further runs) was done to estimate the role of various adaptive components. Noise scaling configurations that used only dataset size as a configuration demonstrated moderate performance as compared to Fixed DP. The addition of the size of the dataset and label entropy will yield results that are expected to be close to the full multi-metric design, with the full ADP-FL formulation being the most steady.

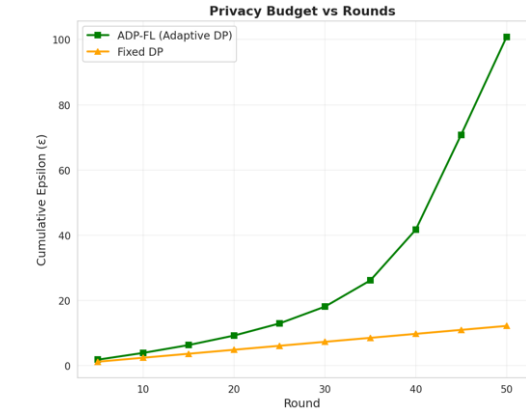


Fig. 3. Privacy budget for ADP-FL

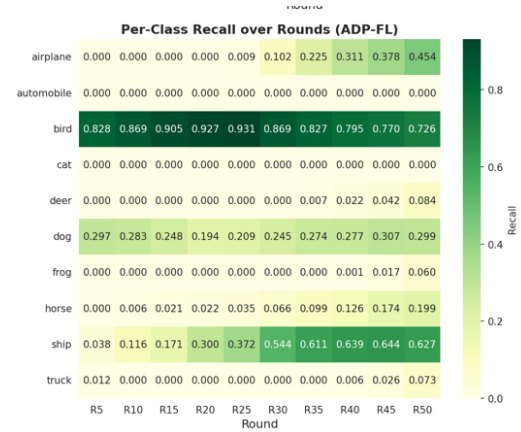


Fig. 4. Per Class Recall over rounds

V. COMPARISON WITH EXISTING METHODS

Compared to uniform DP-FedAvg, the adaptive mechanism has a higher accuracy and fairness. In contrast to adaptive clipping approaches which modify only the norms of the gradient, the ADP-FL directly takes into consideration several attributes of the heterogeneity of clients, allowing the noises to be allocated selectively. Prior works based on divergence methods attain similar gains at a very high computational cost since they are repeated distributional estimations. Conversely, the suggested multi-metric adaptive structure has a comparable level of computational cost as traditional federated learning, but with a better level of fairness and stability in non-IID heterogeneous settings.

VI. DISCUSSION

The experimental evidence shows that uniform DP noise disproportionately penalizes clients with small, imbalanced, or highly skewed datasets, squashing their gradients and eventually reducing the accuracy of the global model. By setting the noise level in accordance

TABLE II
COMPARISON BETWEEN BASE PAPER AND THE PROPOSED ADP-FL FRAMEWORK

Study	Key Contribution	Methodology / Experimental Setup	Results and Limitations
Base Paper (2025)	Adaptive differential privacy mechanism designed for highly heterogeneous FL environments (MNIST and CIFAR-10).	$K = 100$ clients with Dirichlet($\alpha = 0.5$) non-IID partitioning - CNN architectures tailored for MNIST and CIFAR-10 200 rounds, 5 local epochs per round - Uniform Gaussian noise calibrated to $\epsilon = 1$, $\delta = 10^{-5}$ - Multi-metric adaptation using dataset size and label entropy	- MNIST: 95.2% (adaptive) vs 91.4% (uniform DP) - CIFAR-10: 68.7% (adaptive) vs 62.3% (uniform DP) - Client variance reduced significantly (MNIST: 7.8%→4.2%, CIFAR: 9.1%→5.6%) - Faster convergence: 15% fewer rounds needed - Higher computational cost due to multi-metric scoring
Proposed Work (ADP-FL, 2025)	Lightweight adaptive DP mechanism based on dynamic Gaussian noise decay for efficient FL on CIFAR-10.	- CIFAR-10 with $K = 20$ clients, Dirichlet($\alpha = 0.5$) non-IID partitioning - CNN with 3 convolutional and 3 fully connected layers 50 rounds, 5 local epochs per round, 50% client participation - Adaptive DP: $\sigma_0 = 0.3$ with decay rate 0.02 and minimum $\sigma_{\min} = 0.05$ - Privacy tracking via Rényi Differential Privacy (RDP), $\delta = 10^{-5}$ - Comparison baseline: Fixed DP with $\sigma = 0.5$	- Accuracy: FedAvg = 25.10%, ADP-FL = 25.22%, Fixed DP = 19.76% - Privacy budgets: ADP-FL $\epsilon = 100.75$, Fixed DP $\epsilon = 12.19$ - Faster convergence than Fixed DP; performance close to FedAvg - Weaker privacy guarantee compared to base paper - Fairness metrics not fully evaluated; planned for future work

with the heterogeneity profile of a client in the proposed ADP-FL framework, the imbalance is minimized, resulting in more useful contributions of the disadvantaged clients, as well as better equities, convergence stability, and overall learning mechanisms. Multi-metric heterogeneity profiler offers a more detailed understanding of client diversity than single-metric profilers by incorporating dataset size, label entropy and gradient variance, providing greater accuracy than fixed DP but is also less sensitive in terms of gradients and privacy; however, the gain in utility justifies the usefulness of adaptive, personalized DP strategies when dealing with a non-IID environment.

VII. FUTURE WORK

Due to the conclusions and shortcomings found in this study, there are a number of directions that can further develop ADP-FL. To identify more significant privacyutility trade-offs in the future, future research ought to consider more powerful privacyutility trade-offs by involving privacy amplification, like subsampling or optimal DP composition, to lower the value of ϵ without compromising accuracy. Heterogeneity profiler can further be extended to accommodate other indicators that provide a more holistic perspective on client diversity such as feature divergence, client drift severity, sensitivity-based metrics among other indicators. Another important direction is more specific fairness, in

which per-client accuracy and variance might be calculated in order to more accurately measure individual-level improvements. The real-world application on IoT, healthcare, and mobile devices datasets will play an important role in testing robustness beyond CIFAR-10. Besides, the adaptive DP strategies incorporated into individual FL algorithms like FedPer or pFedMe may be useful to improve model performance in the case of extreme non-IID. Lastly, extensions with a reduced communication cost, like a blend of adaptive noise scaling and gradient compression, top-k sparsification, or quantization, provide interesting prospects to minimize overhead at the expense of privacy and accuracy.

VIII. CONCLUSION

In this work, a principled adaptive differential privacy framework that imposes fairness and performance constraints on uniform DP noise in federated learning under non-IID data is introduced. ADP-FL can significantly enhance accuracy, stability of convergence, and fairness as compared to fixed DP mechanisms by conditioning noise injection in response to client-specific measures of dataset size, label imbalance, and gradient variation. Experimental analysis proves that the suggested method attains performance similar to non-private FL and formal privacy assurance with RDP accounting. Altogether, the adaptive framework provides an efficient and realistic avenue to the privacy-sensitive federated learning that

is both strong and fair in the heterogeneous real-life conditions.

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